Kemagnetan (Gaya Lorentz)

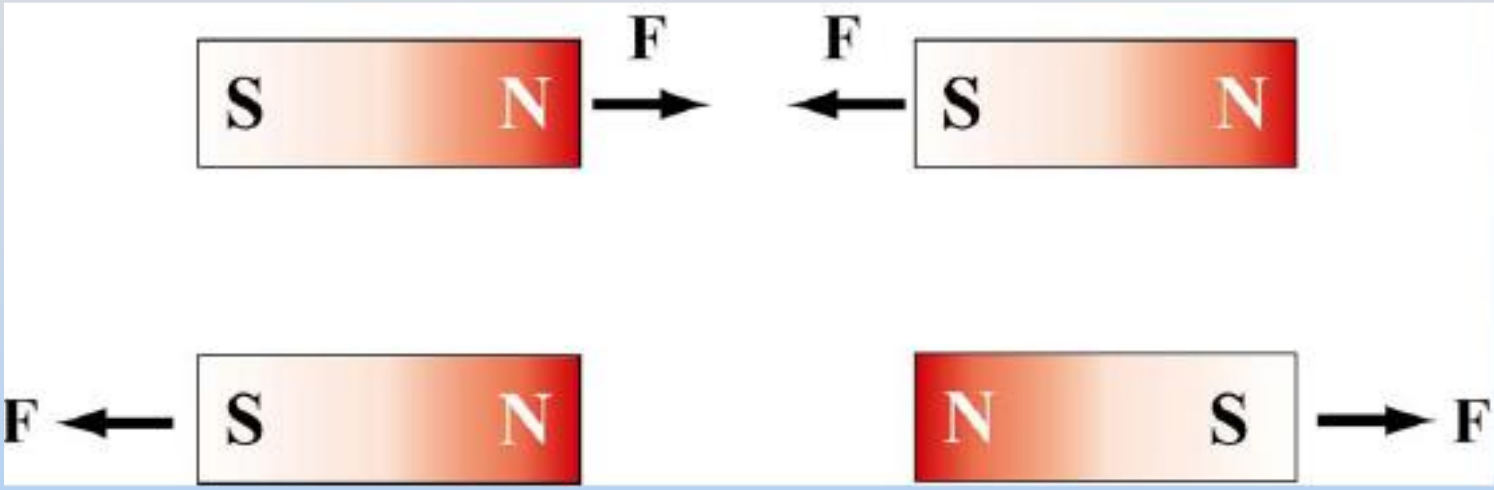
Dosen : Defrianto Pratama, S.Pd, M.Si.

Jurusan Teknik Elektro

Politeknik Negeri Bandung

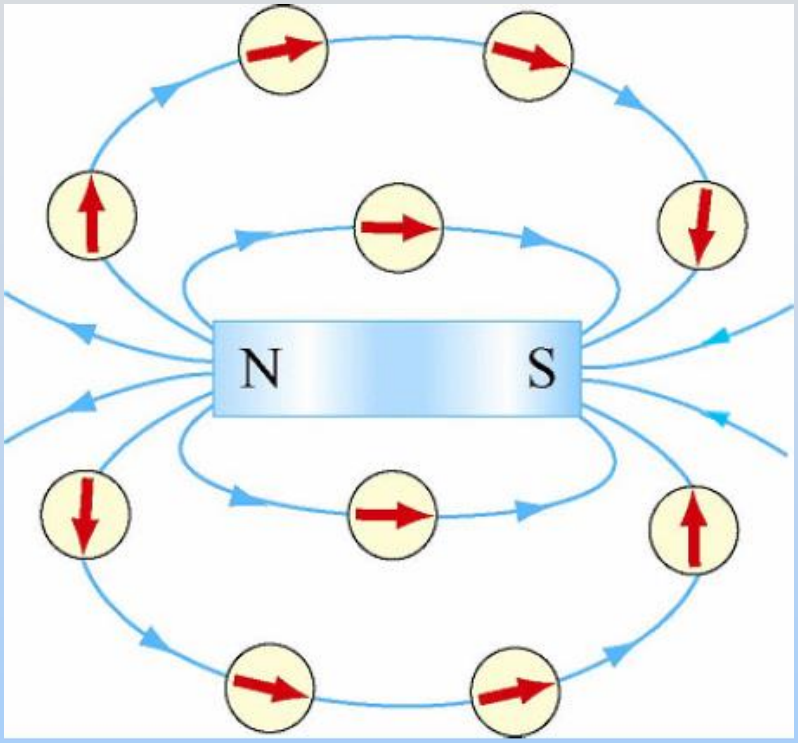
Sumber bacaan Ebook: Halliday ch 28

Magnet Batang



Kutub sejenis tolak menolak, berlawanan jenis tarik menarik

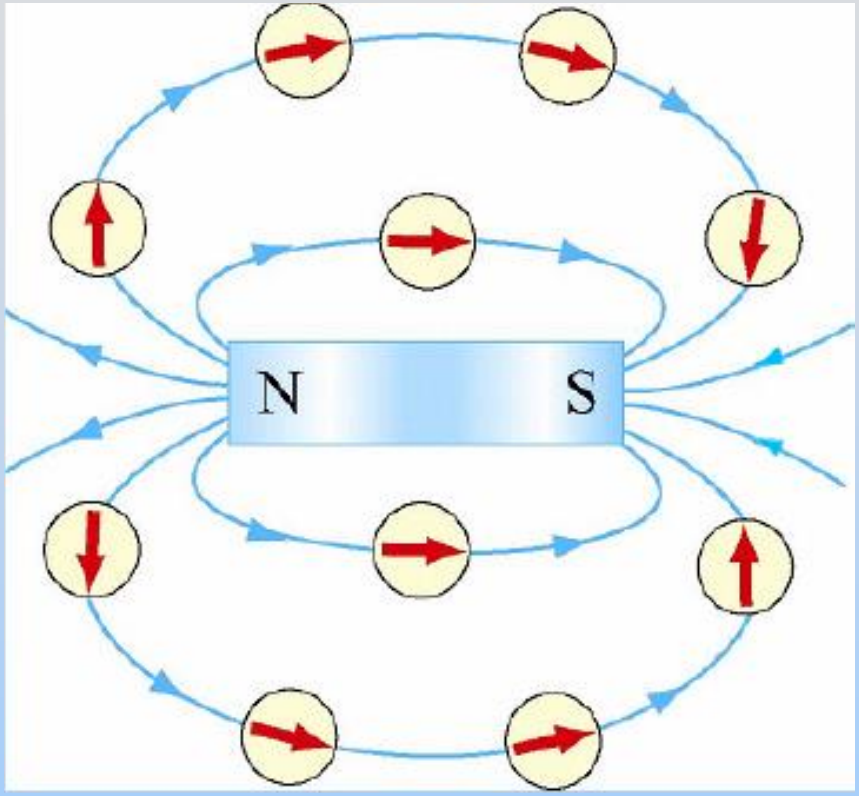
Medan Magnet dari Magnet Batang



(1) Sebuah magnet mempunyai dua kutub, North (N) dan South (S)

(2) Arah medan Magnet meninggalkan N, berakhir di S

Magnet Batang adalah Dipol!



Menghasilkan medan dipol

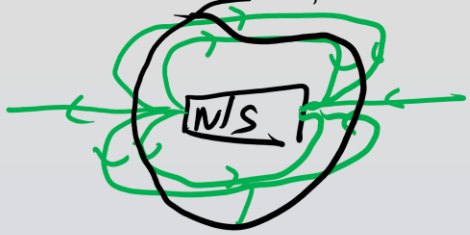
Apakah terdapat “massa” Magnet atau “muatan” Magnet?





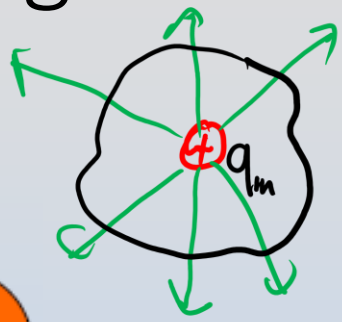
Tidak! Monopol magnet tidak ditemukan sendirian

Monopol Magnet ?



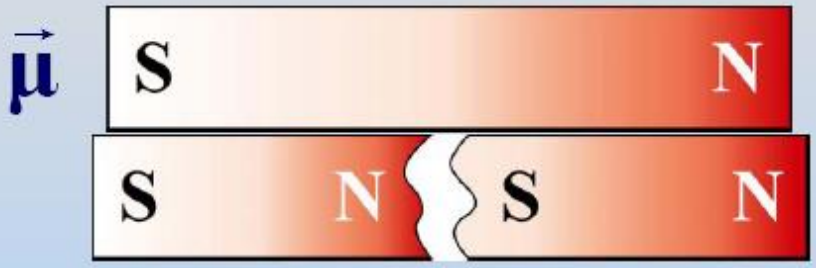
Dipol Magnet

Dipol Listrik





Ketika dipotong:
2 monopol (muatan)



Ketika dipotong: 2 dipol

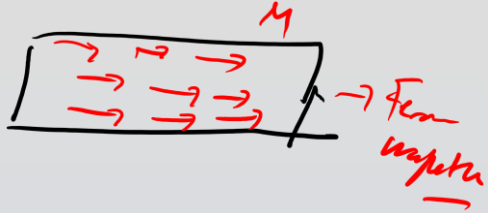
Monopol magnet tidak ditemukan sendirian
Persamaan Maxwell yang lain (2 dari 4)

$$\oiint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{A}} = \frac{q_{in}}{\varepsilon_0}$$

Hukum Gauss

$$\oiint_S \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = 0$$

Hukum Gauss Magnet



Medan: Grav., Listrik, Magnet

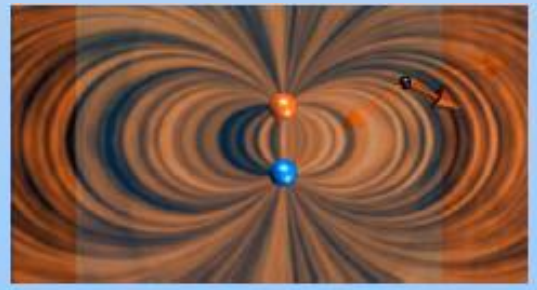
Massa *m* Muatan *q* (±)

Menghasilkan: $\vec{\mathbf{g}} = -G\frac{m}{r^2}\hat{\mathbf{r}}$ $\vec{\mathbf{E}} = k_e\frac{q}{r^2}\hat{\mathbf{r}}$

Tidak ada Monopole Magnet

Merasakan: $\vec{\mathbf{F}}_g = m\vec{\mathbf{g}}$ $\vec{\mathbf{F}}_E = q\vec{\mathbf{E}}$

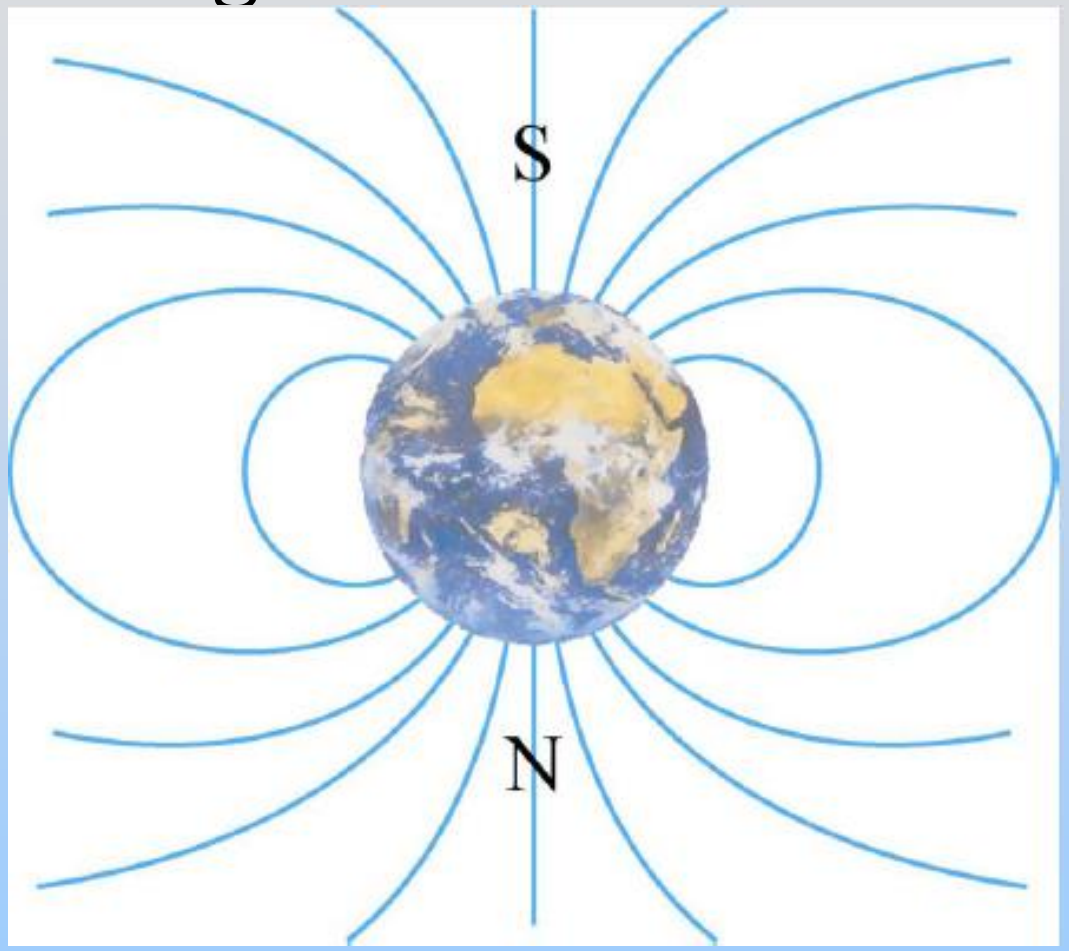
Tinjau juga… Dipol **p** Dipol μ

Menghasilkan: $\vec{\mathbf{E}} \rightarrow$  $\leftarrow \vec{\mathbf{B}}$

Merasakan: $\vec{\boldsymbol{\tau}} = \vec{\mathbf{p}} \times \vec{\mathbf{E}}$ $\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{\mu}} \times \vec{\mathbf{B}}$

Apa lagi yang termasuk magnet?

Medan Magnet Bumi

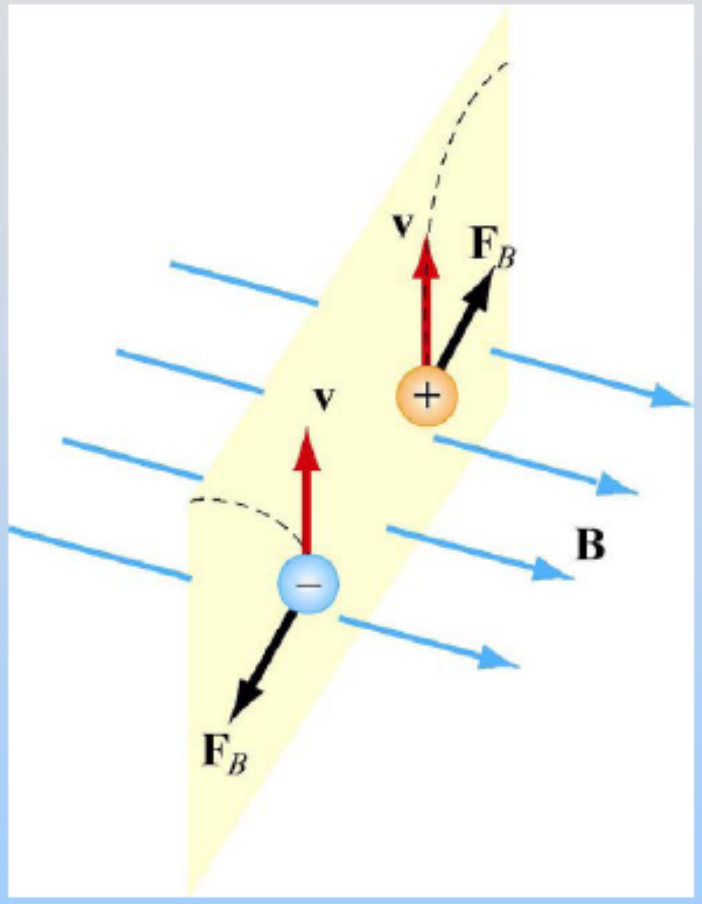


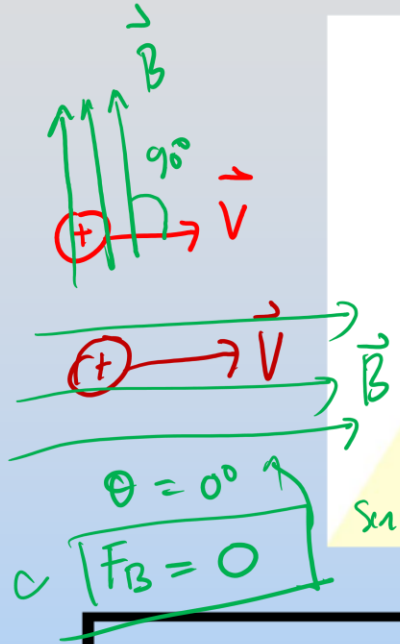
Juga merupakan dipol magnet

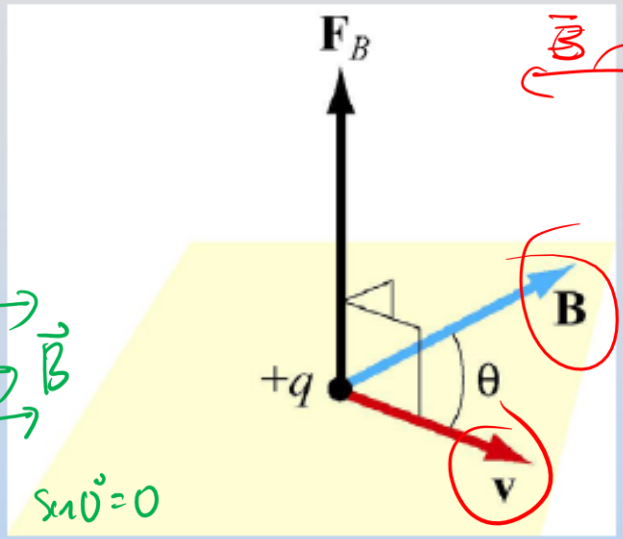
Kutub utara magnet adalah kutub selatan bumi

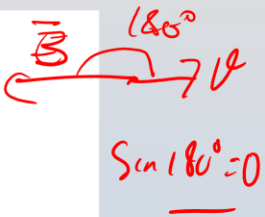
- Gaya Magnet

Muatan yang Bergerak Merasakan Gaya Magnet









$$\vec{F}_B = q\vec{v} \times \vec{B}$$

Gaya magnet tegak lurus terhadap:
Kecepatan **v** dari muatan dan medan magnet **B**

$F_B = qvB \sin\theta$

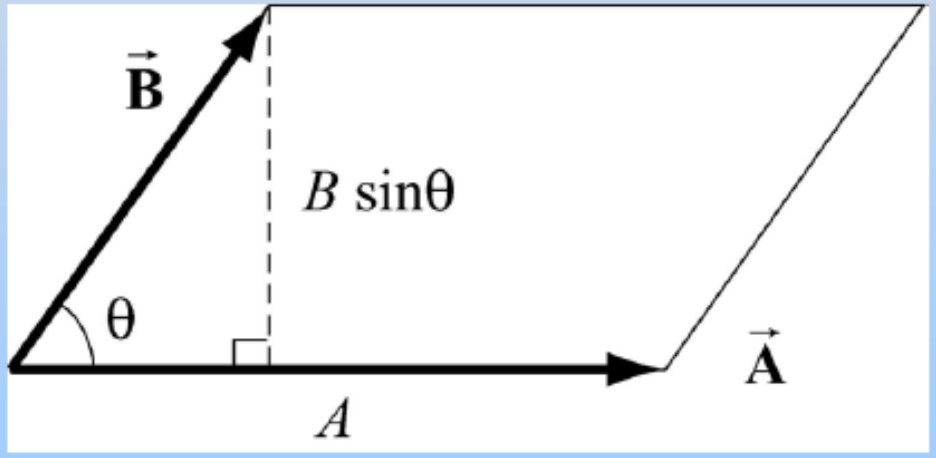
$F_{B\,max} = qvB \rightarrow \theta = 90^\circ$

Recall:
Cross Product

Cross Product: Besarnya

Menghitung besarnya cross product A x B:

$$\vec{C} = \vec{A} \times \vec{B} \qquad \left|\vec{C}\right| = \left|\vec{A}\right|\left|\vec{B}\right| \sin\theta$$



$|\vec{C}|$: Luas Trapesium

Medan Magnet B: Satuan

Karena $$\vec{\mathbf{F}}_B = q\,\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

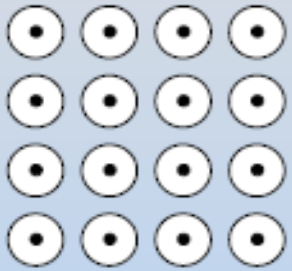
$$\text{B Units} = \frac{\text{newton}}{(\text{coulomb})(\text{meter/second})} = 1\frac{\text{N}}{\text{C}\cdot\text{m/s}} = 1\frac{\text{N}}{\text{A}\cdot\text{m}}$$

This is called 1 Tesla (T)

$B_{\text{Bumi}} \approx 25-65\ \mu\text{T}$

$$1\ \text{T} = 10^4\ \text{Gauss (G)}$$

Notasi

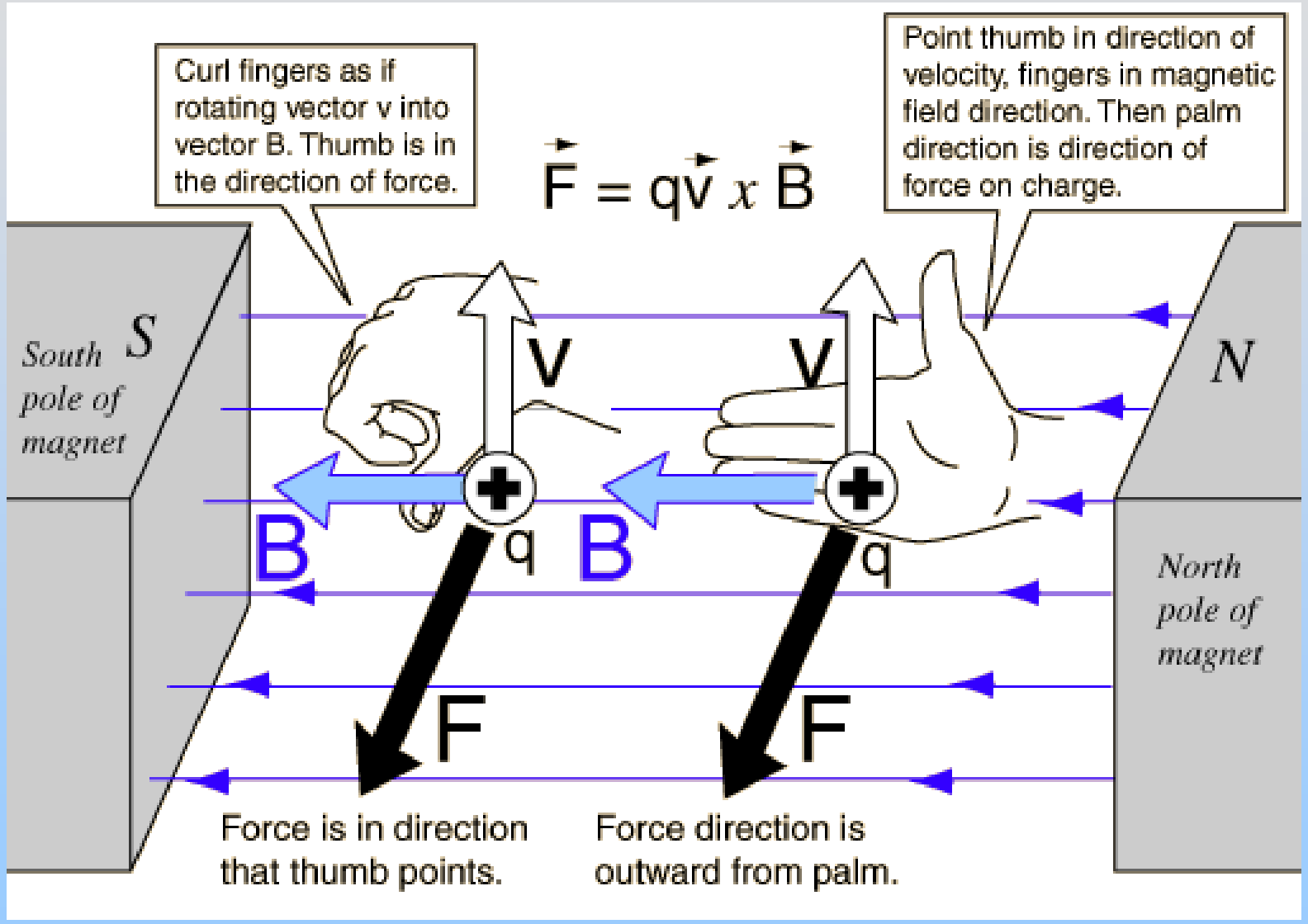


Keluar bidang
"kepala anak panah"



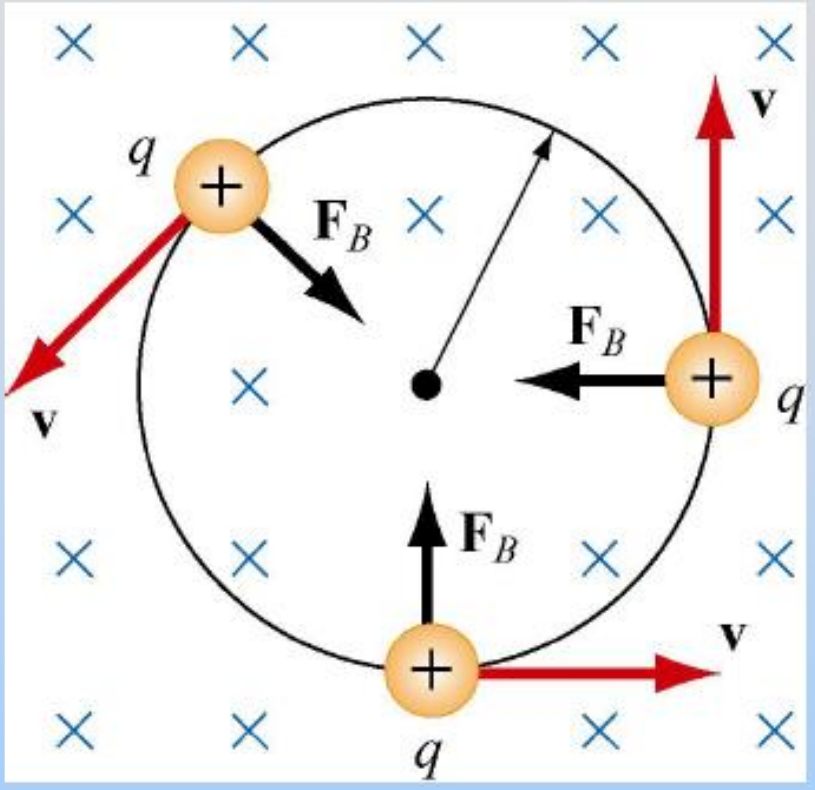
Masuk bidang
"ujung anak panah"

Aturan Tangan Kanan 1



- Jenis Gerak Seperti Apa yang Dihasilkan?

Gerak Siklotron



(1) r : Jari-jari lingkaran

$$qvB = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB}$$

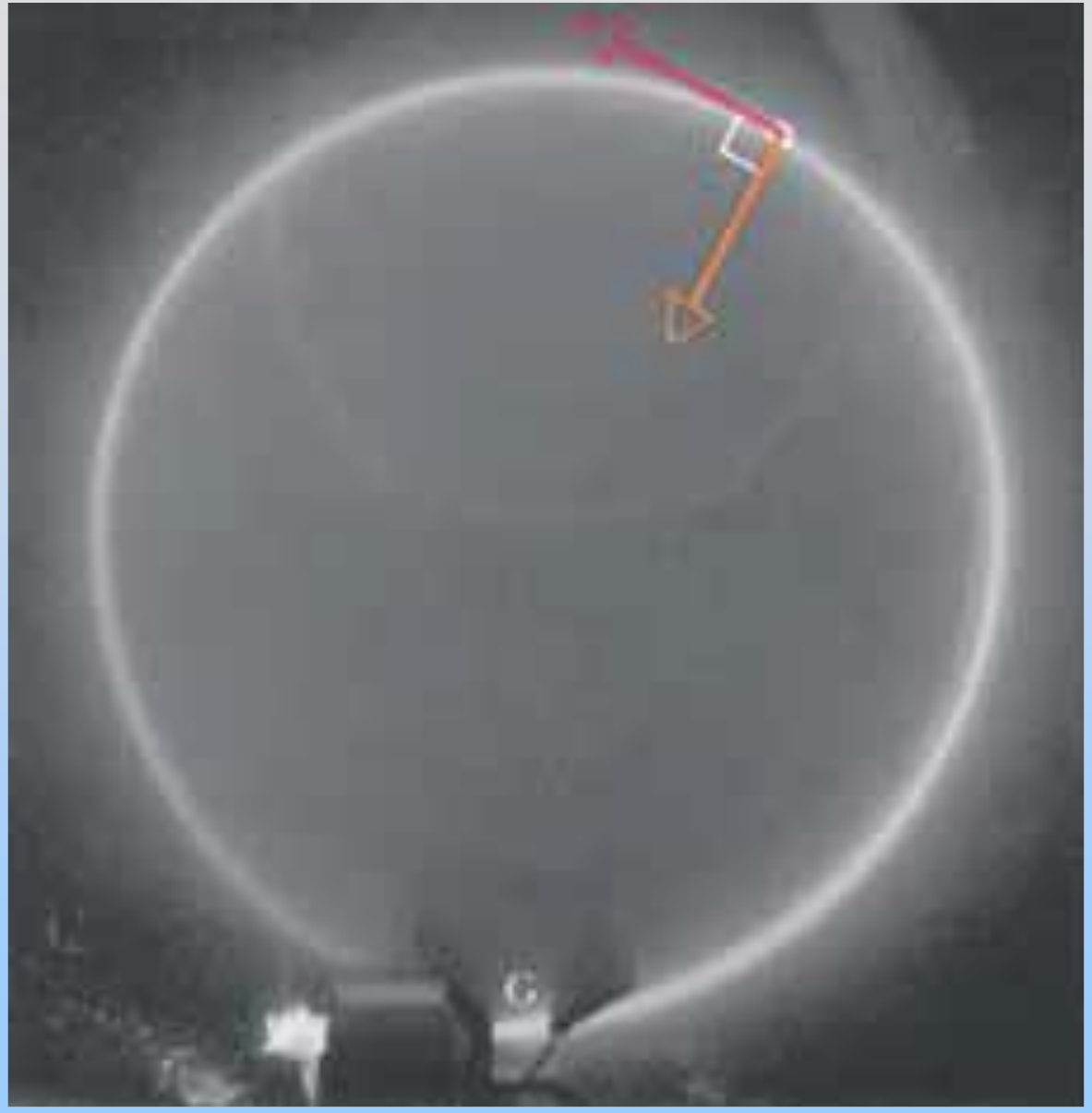
(2) T : periode gerak

$$T = \frac{2\pi r}{v} = \frac{2\pi m}{qB}$$

(3) ω : frekuensi siklotron

$$\omega = 2\pi f = \frac{v}{r} = \frac{qB}{m}$$

Animasi 7.2 Animasi 7.3



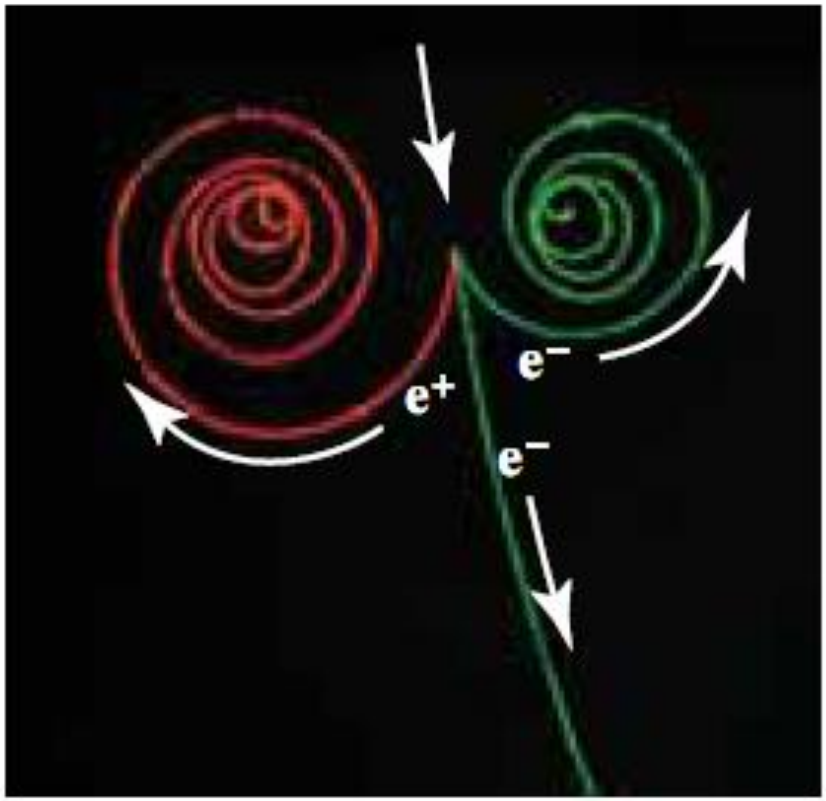


Fig. 28-3 The tracks of two electrons (e^-) and a positron (e^+) in a bubble chamber that is immersed in a uniform magnetic field that is directed out of the plane of the page. *(Lawrence Berkeley Laboratory/Photo Researchers)*

Gaya Magnet + Gaya Listrik: Gaya Lorentz

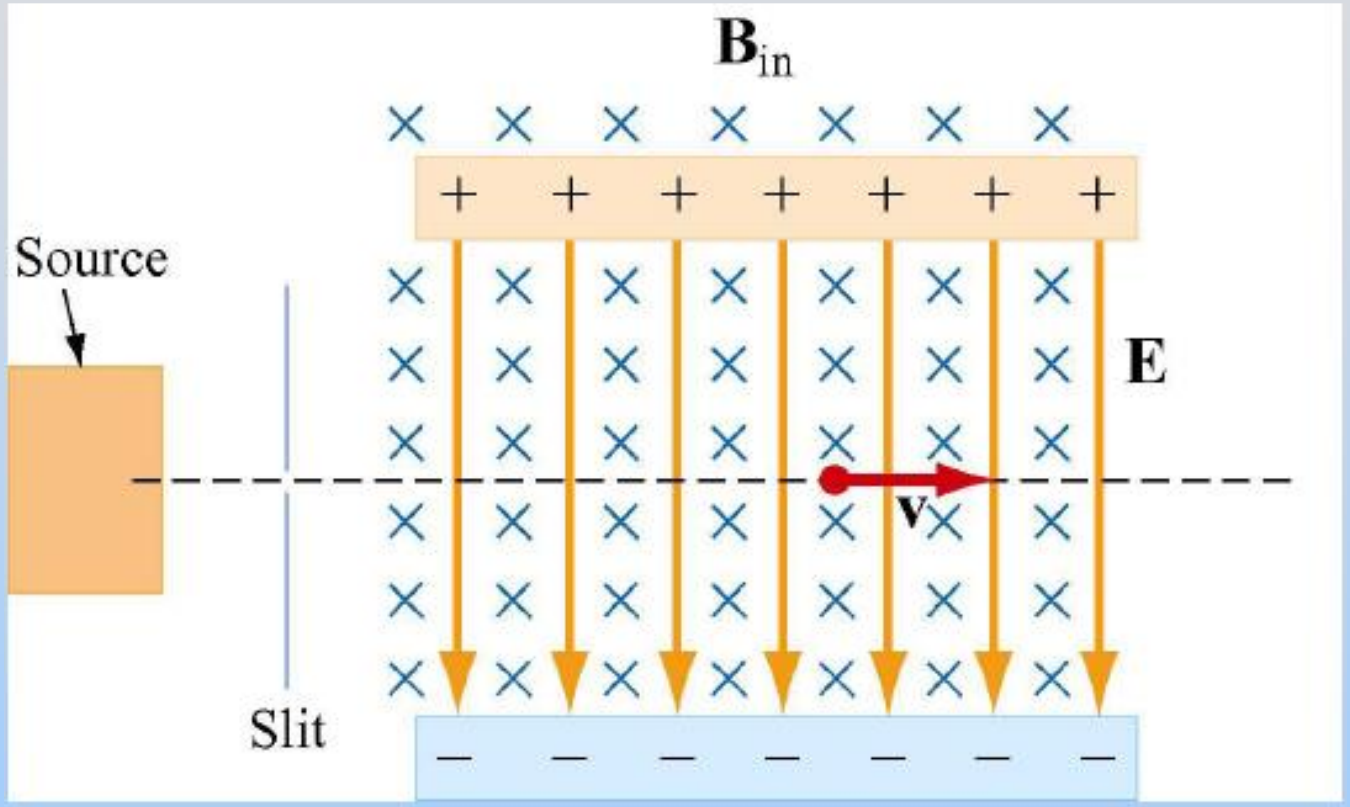
Muatan merasakan…

$$\vec{\mathbf{F}}_E = q\vec{\mathbf{E}} \qquad \vec{\mathbf{F}}_B = q\,\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

Medan Listrik Medan Magnet

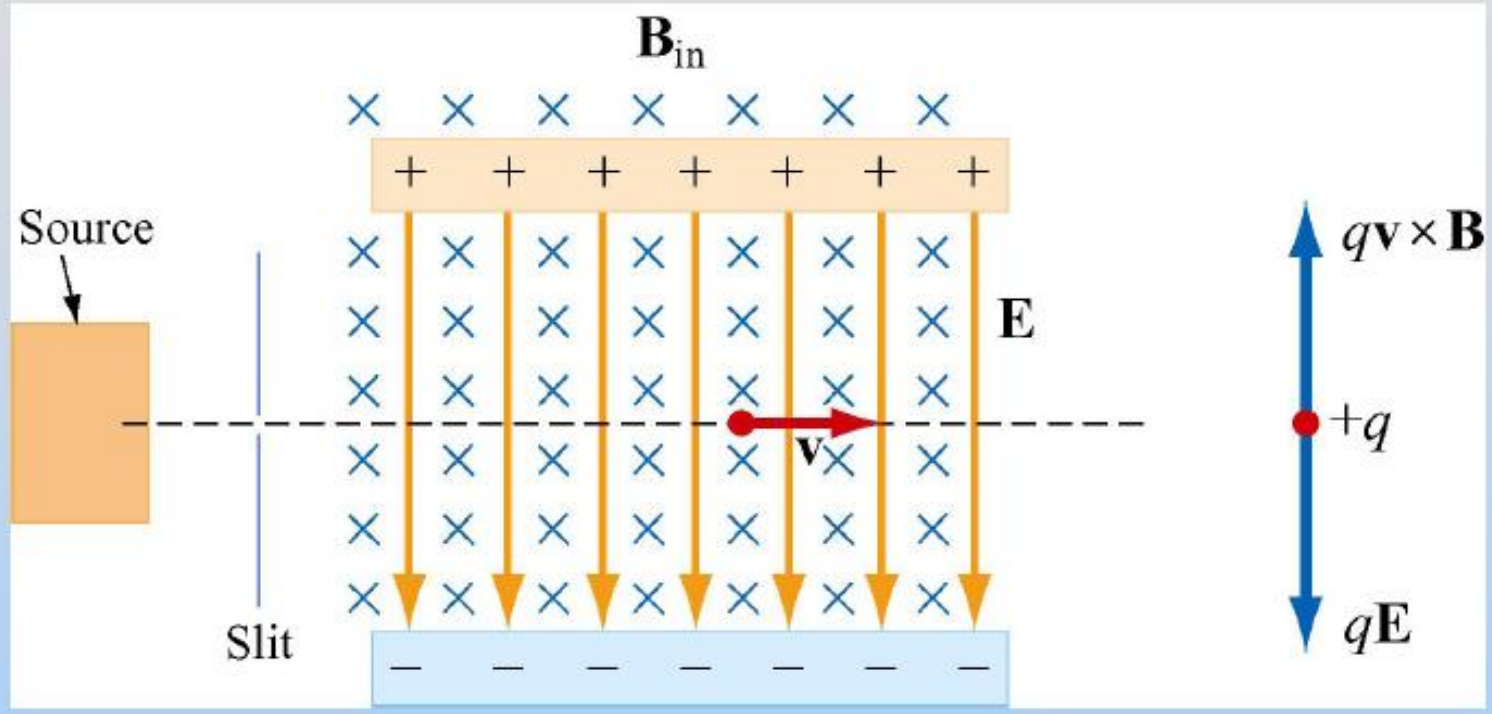
$$\boxed{\vec{\mathbf{F}} = q\left(\vec{\mathbf{E}} + \vec{\mathbf{v}} \times \vec{\mathbf{B}}\right)}$$

Aplikasi: Selektor Kecepatan



Apa yang terjadi disini?

Selektor Kecepatan



Partikel bergerak sepanjang garis lurus ketika:

$$F_B = F_E$$

$$qvB = qE$$

$$v = \frac{E}{B}$$

$$\mathbf{F}_{net} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B}) = 0 \Rightarrow v = \frac{E}{B}$$

Lintasan Helix

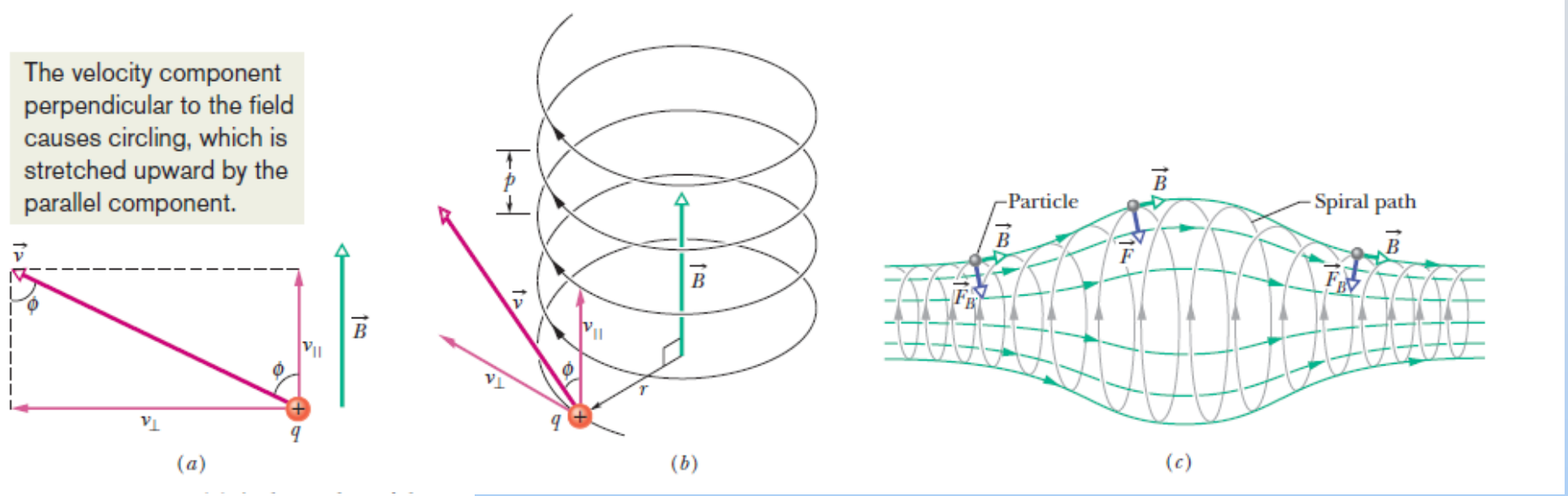
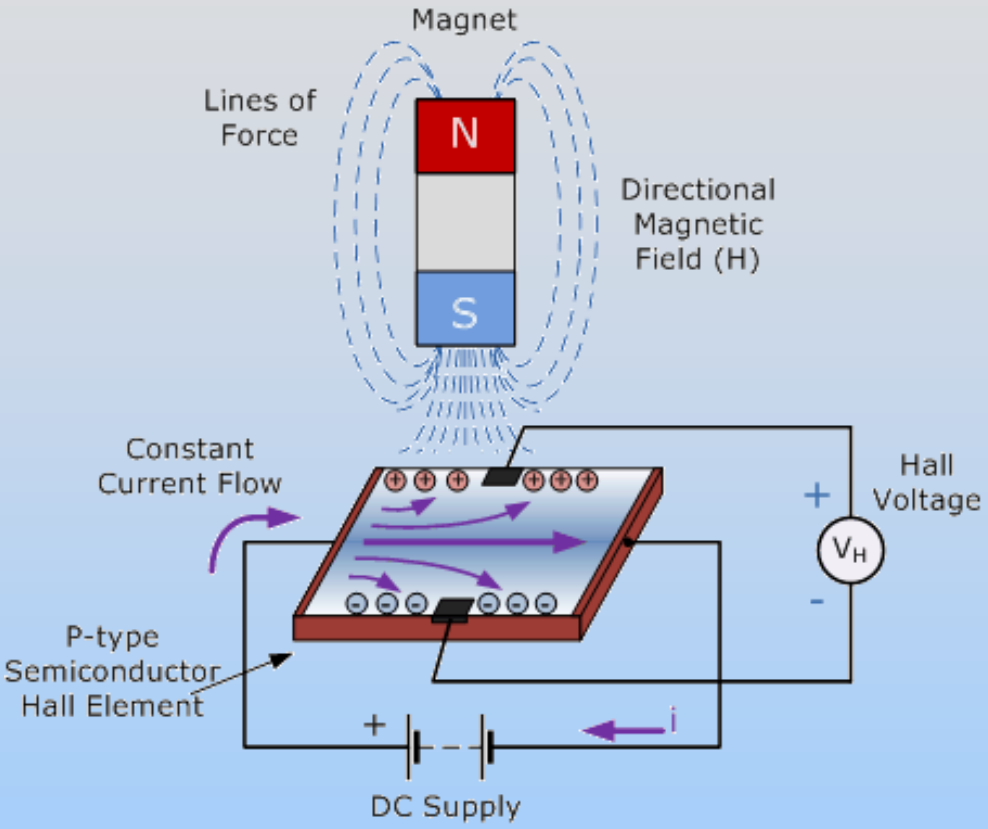
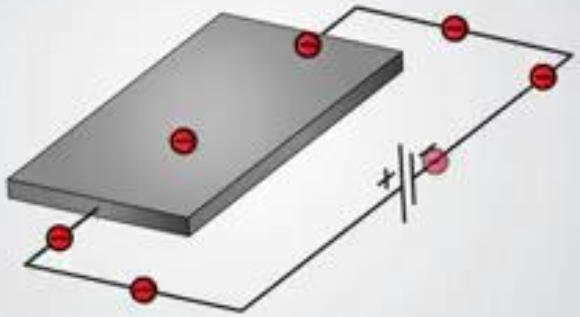


Fig. 28-11 (*a*) A charged particle moves in a uniform magnetic field $\vec{B}$, the particle's velocity $\vec{v}$ making an angle ϕ with the field direction. (*b*) The particle follows a helical path of radius r and pitch p. (*c*) A charged particle spiraling in a nonuniform magnetic field. (The particle can become trapped, spiraling back and forth between the strong field regions at either end.) Note that the magnetic force vectors at the left and right sides have a component pointing toward the center of the figure.

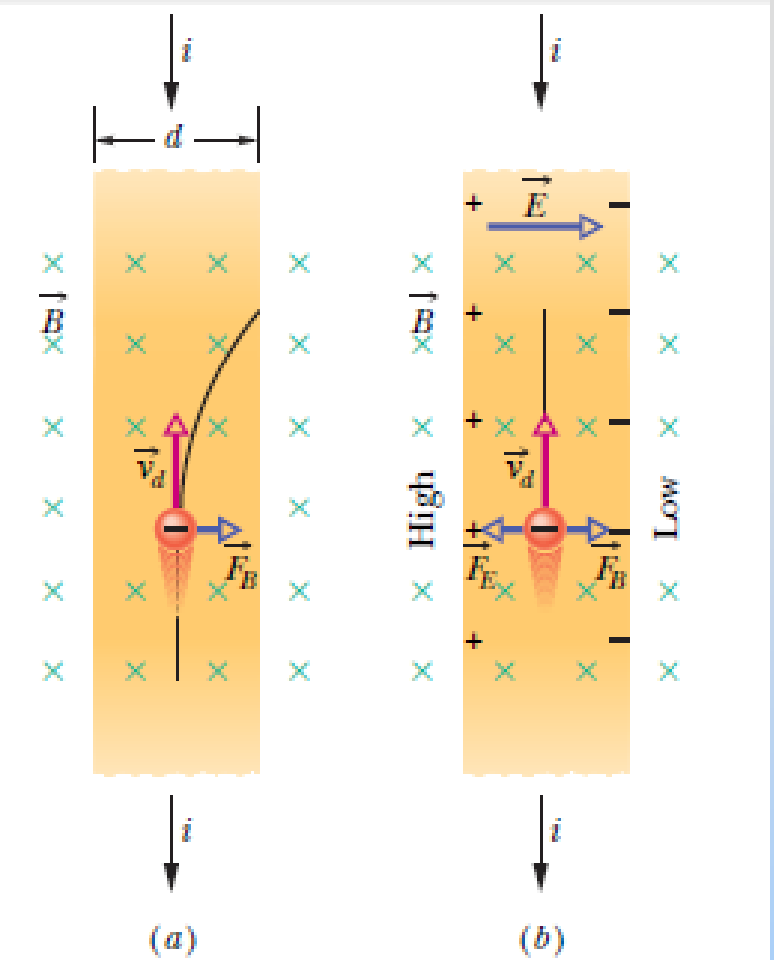
Efek Hall


Magnet
Lines of Force
N
S
Directional Magnetic Field (H)
Constant Current Flow
Hall Voltage
+
V_H
-
P-type Semiconductor Hall Element
+
i
DC Supply

Hall Effect



How To Mechatronics
www.HowToMechatronics.com

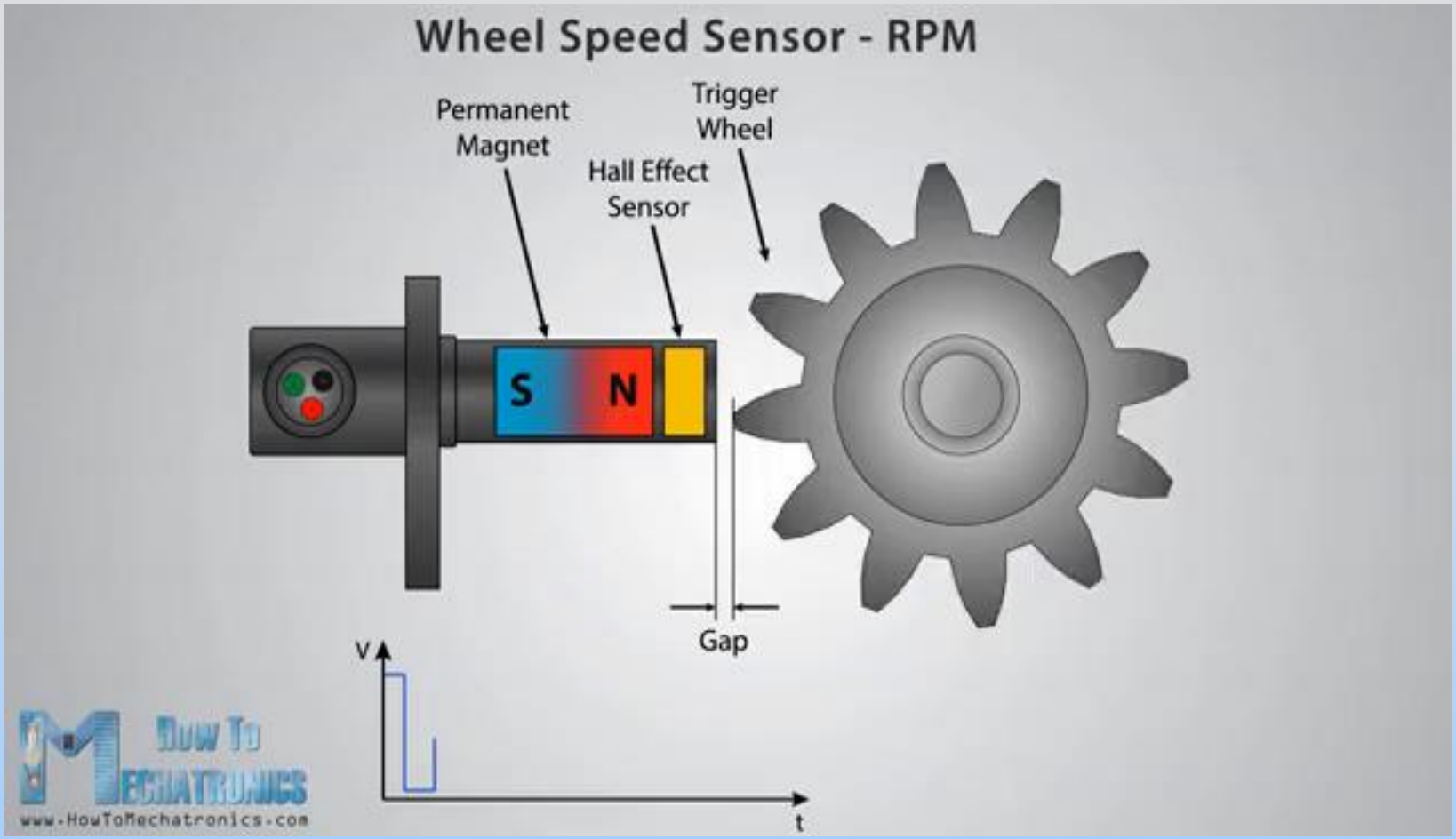


$$V = Ed.$$

$$eE = ev_d B.$$

$$v_d = \frac{J}{ne} = \frac{i}{neA},$$

$$n = \frac{Bi}{Vle},$$


Wheel Speed Sensor - RPM
Permanent Magnet
Trigger Wheel
Hall Effect Sensor
S
N
Gap
V
t
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- **Kawat Berarus**

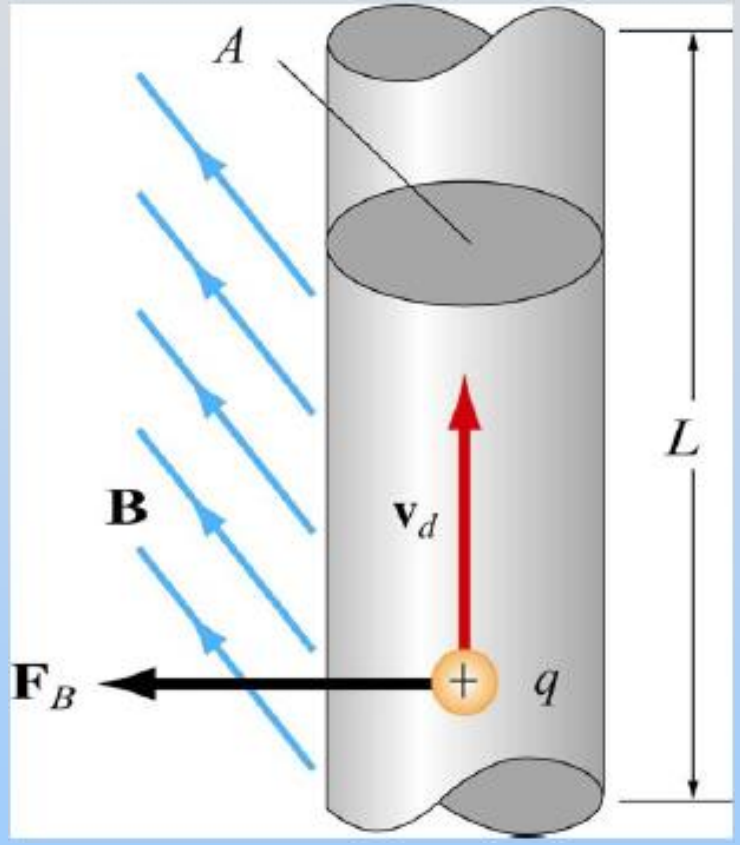
Gaya Magnet pada Kawat Berarus



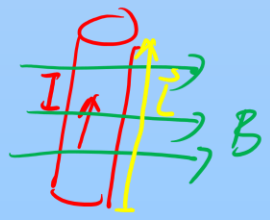
$$\vec{\mathbf{F}}_B = q\vec{\mathbf{v}} \times \vec{\mathbf{B}}$$

$$= (\text{charge})\frac{\text{m}}{\text{s}} \times \vec{\mathbf{B}}$$

$$= \frac{\text{charge}}{\text{s}}\text{m} \times \vec{\mathbf{B}}$$

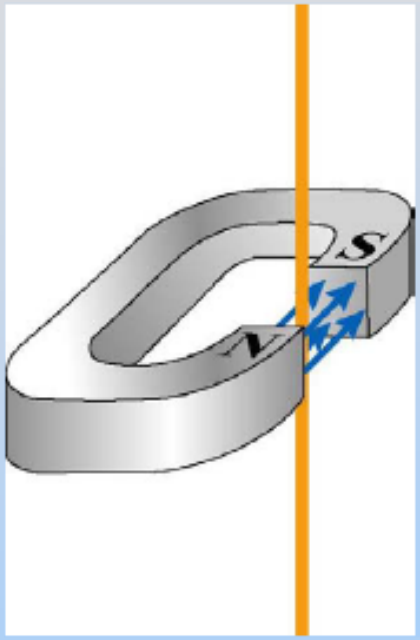


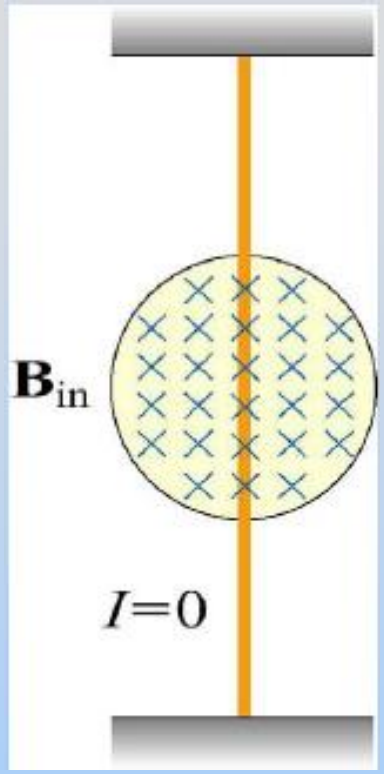
$$\boxed{\vec{\mathbf{F}}_B = I\left(\vec{\mathbf{L}} \times \vec{\mathbf{B}}\right)}$$

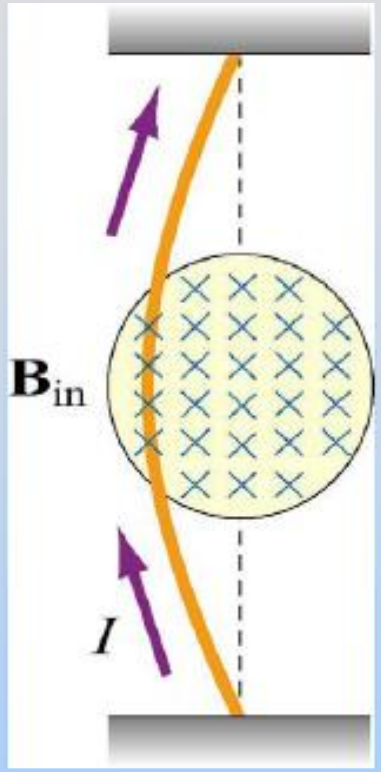


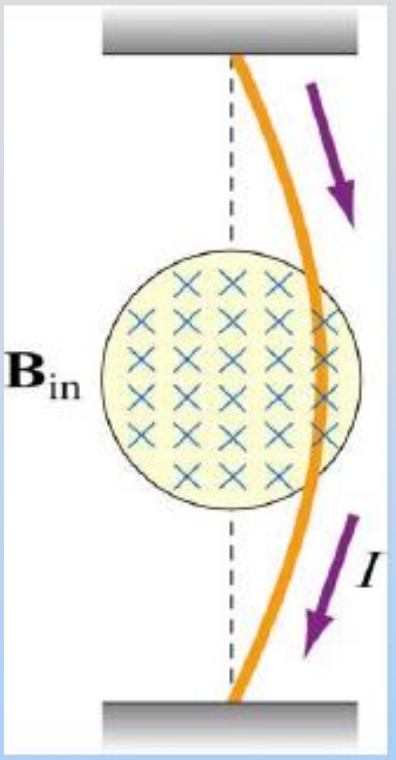
$F_{B\,max} = ILB$

Gaya Magnet pada Kawat Berarus









Arus adalah muatan yang bergerak, dan kita ketahui bahwa muatan yang bergerak **merasakan** gaya dalam suatu medan magnet

Animasi 7.8

Sample Problem

Magnetic force on a wire carrying current

A straight, horizontal length of copper wire has a current $i = 28$ A through it. What are the magnitude and direction of the minimum magnetic field $\vec{B}$ needed to suspend the wire—that is, to balance the gravitational force on it? The linear density (mass per unit length) of the wire is 46.6 g/m.

KEY IDEAS

(1) Because the wire carries a current, a magnetic force $\vec{F}_B$ can act on the wire if we place it in a magnetic field $\vec{B}$. To balance the downward gravitational force $\vec{F}_g$ on the wire, we want $\vec{F}_B$ to be directed upward (Fig. 28-17). (2) The direction of $\vec{F}_B$ is related to the directions of $\vec{B}$ and the wire's length vector $\vec{L}$ by Eq. 28-26 ($\vec{F}_B = i\vec{L} \times \vec{B}$).

Calculations: Because $\vec{L}$ is directed horizontally (and the current is taken to be positive), Eq. 28-26 and the right-hand rule for cross products tell us that $\vec{B}$ must be horizontal and rightward (in Fig. 28-17) to give the required upward $\vec{F}_B$.

The magnitude of $\vec{F}_B$ is $F_B = iLB \sin \phi$ (Eq. 28-27). Because we want $\vec{F}_B$ to balance $\vec{F}_g$, we want

$$iLB \sin \phi = mg, \tag{28-29}$$

where mg is the magnitude of $\vec{F}_g$ and m is the mass of the wire.

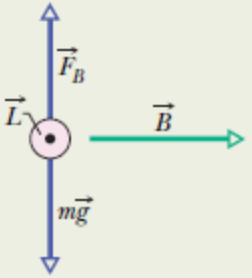


Fig. 28-17 A wire (shown in cross section) carrying current out of the page.

We also want the minimal field magnitude B for $\vec{F}_B$ to balance $\vec{F}_g$. Thus, we need to maximize $\sin \phi$ in Eq. 28-29. To do so, we set $\phi = 90°$, thereby arranging for $\vec{B}$ to be perpendicular to the wire. We then have $\sin \phi = 1$, so Eq. 28-29 yields

$$B = \frac{mg}{iL \sin \phi} = \frac{(m/L)g}{i}. \tag{28-30}$$

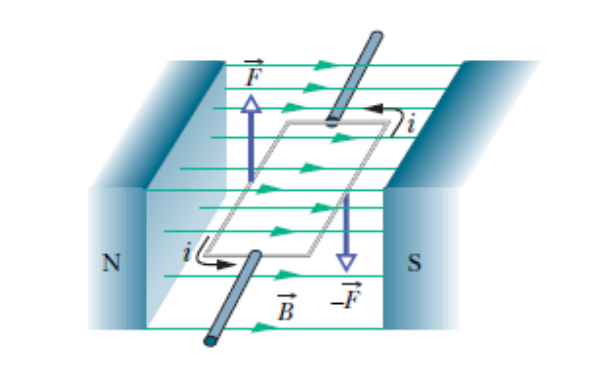
We write the result this way because we know m/L, the linear density of the wire. Substituting known data then gives us

$$B = \frac{(46.6 \times 10^{-3}\ \text{kg/m})(9.8\ \text{m/s}^2)}{28\ \text{A}}$$

$$= 1.6 \times 10^{-2}\ \text{T}. \qquad \text{(Answer)}$$

This is about 160 times the strength of Earth's magnetic field.

Motor Listrik Sederhana



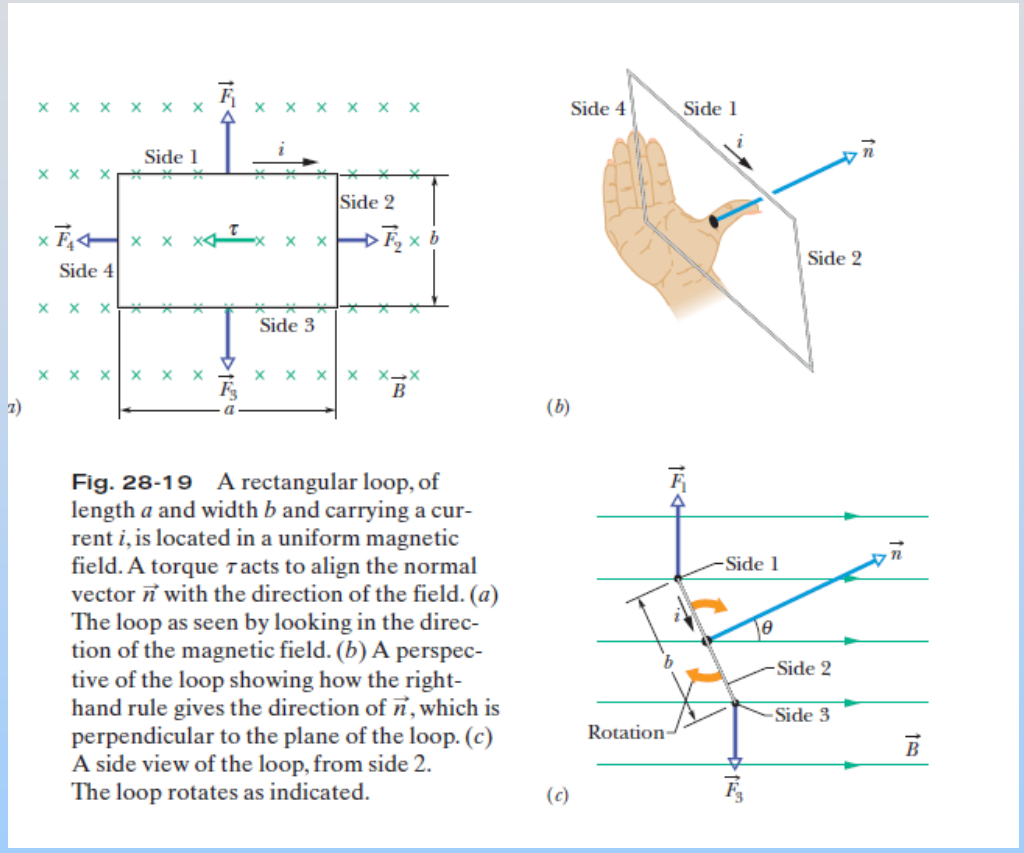


Fig. 28-19 A rectangular loop, of length a and width b and carrying a current i, is located in a uniform magnetic field. A torque τ acts to align the normal vector $\vec{n}$ with the direction of the field. (*a*) The loop as seen by looking in the direction of the magnetic field. (*b*) A perspective of the loop showing how the right-hand rule gives the direction of $\vec{n}$, which is perpendicular to the plane of the loop. (*c*) A side view of the loop, from side 2. The loop rotates as indicated.

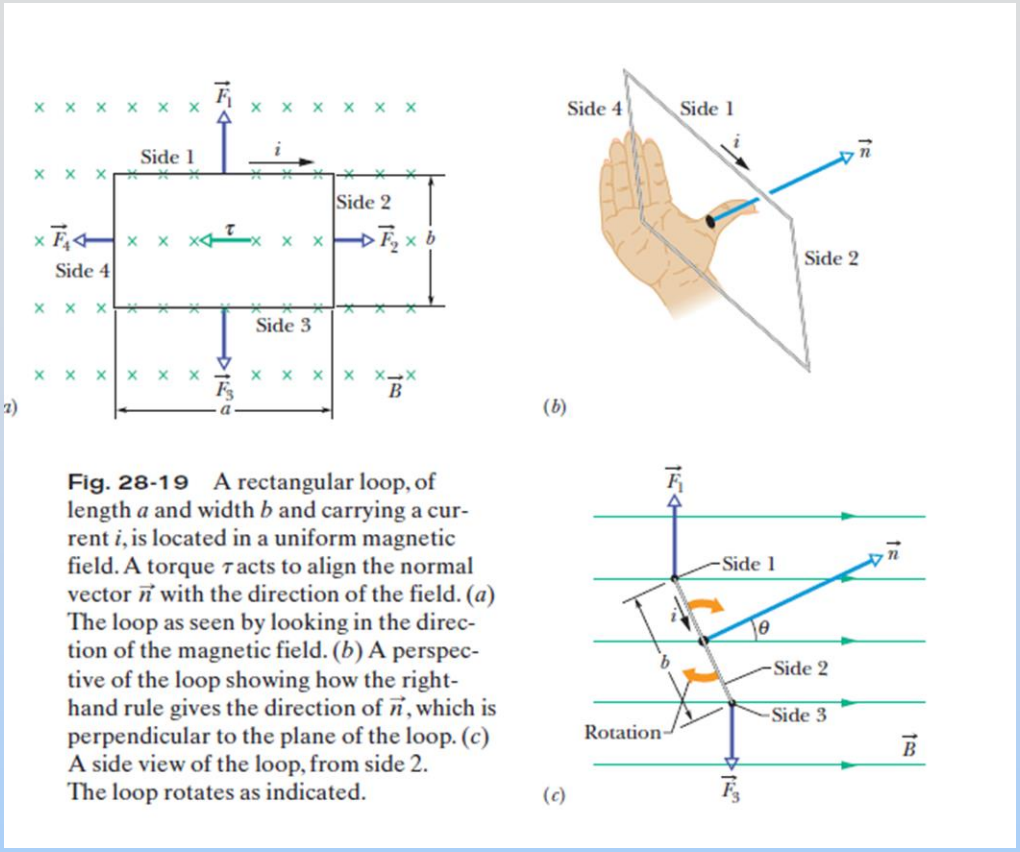


Fig. 28-19 A rectangular loop, of length a and width b and carrying a current i, is located in a uniform magnetic field. A torque τ acts to align the normal vector $\vec{n}$ with the direction of the field. (*a*) The loop as seen by looking in the direction of the magnetic field. (*b*) A perspective of the loop showing how the right-hand rule gives the direction of $\vec{n}$, which is perpendicular to the plane of the loop. (*c*) A side view of the loop, from side 2. The loop rotates as indicated.

$$\tau = Fr \sin\theta$$

$$\tau' = \left(iaB\frac{b}{2}\sin\theta\right) + \left(iaB\frac{b}{2}\sin\theta\right) = iabB\sin\theta.$$

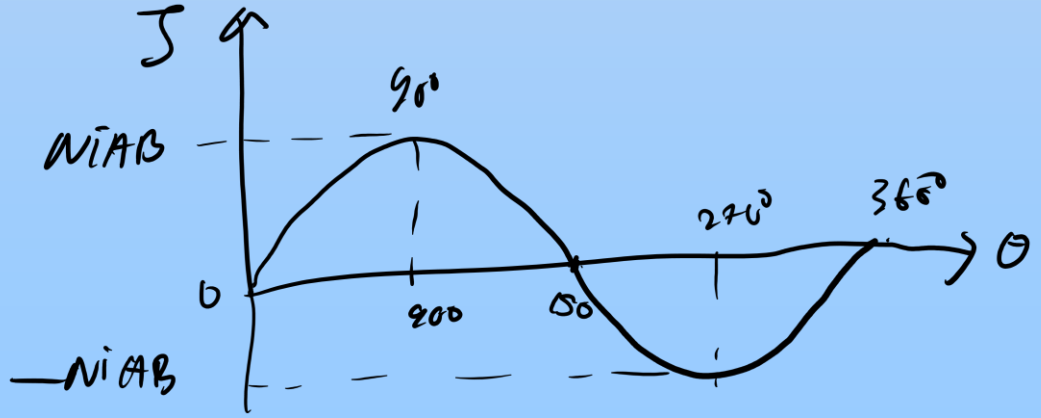
$$\tau = N\tau' = NiabB\sin\theta = (NiA)B\sin\theta,$$

Jika kawat berbentuk Lingkaran

$$\tau = (Ni\pi r^2)B\sin\theta.$$

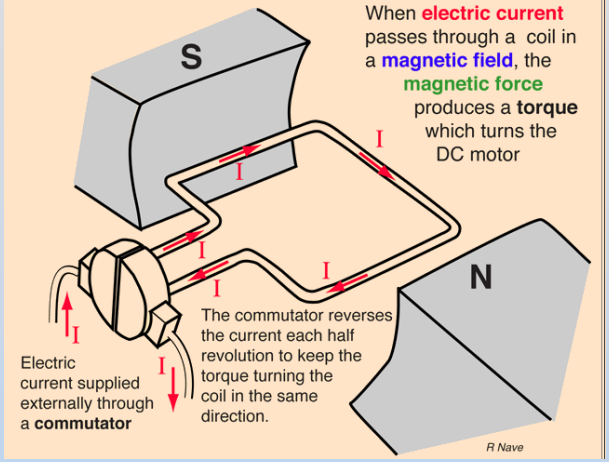
$J = NiAB \sin\theta$

$J_{max} = NiAB$ ← $\theta = 90^\circ$



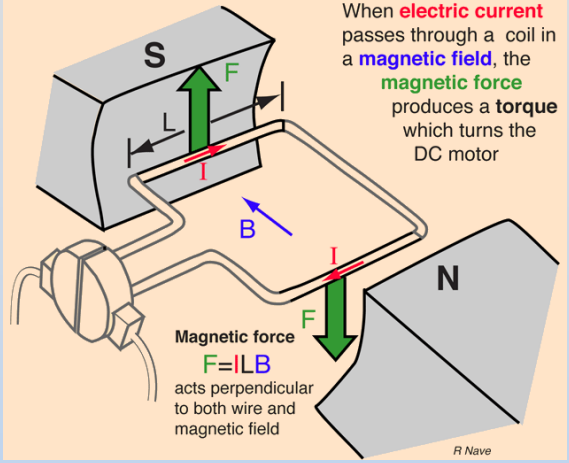
Current in DC Motor

This is an active graphic. Click on bold type for further illustration.



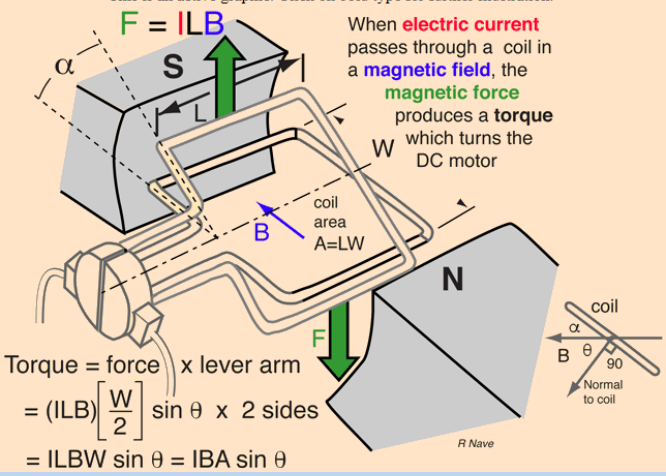
Force in DC Motor

This is an active graphic. Click on bold type for further illustration.

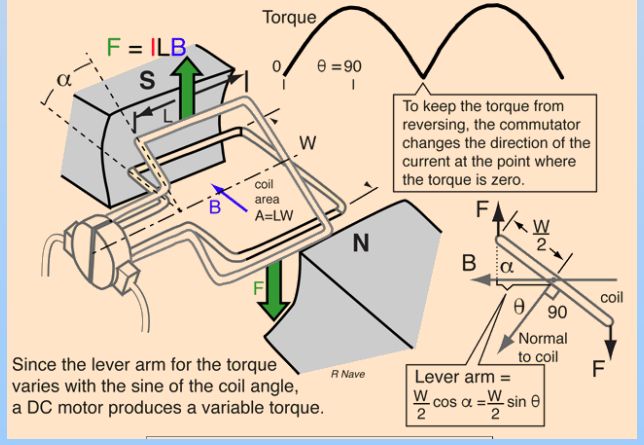


Torque in DC Motor

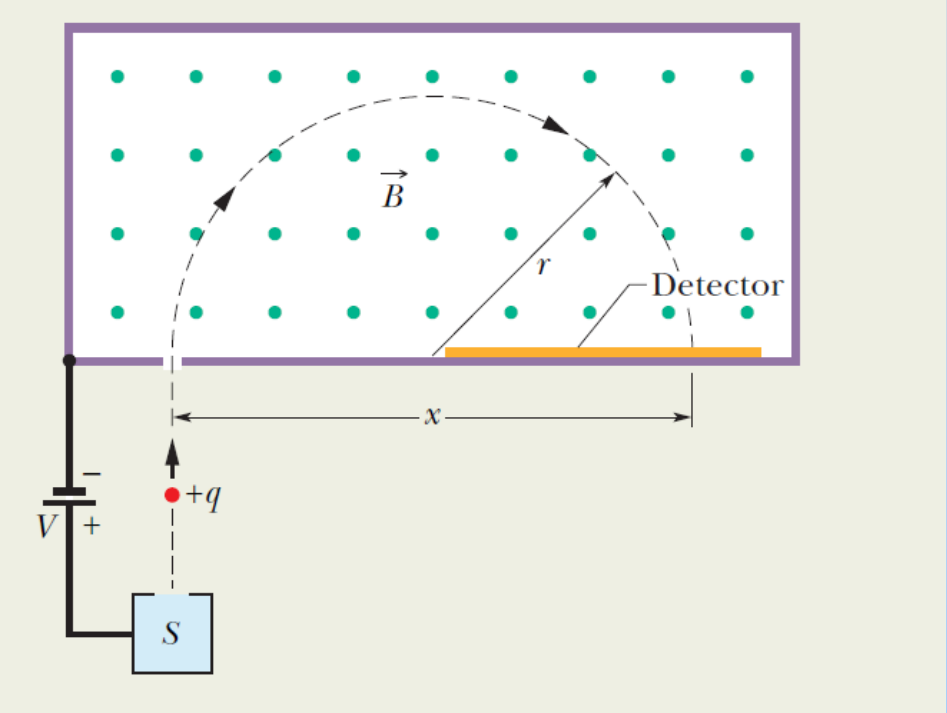
This is an active graphic. Click on bold type for further illustration.



Torque Variation in DC Motor



TUGAS NO 1



Spektometer Massa adalah alat untuk mengukur massa sebuah partikel yang bermuatan. Partikel bermuatan q ditembakan pada sebuah daerah dengan medan magnet B uniform yang tegak lurus dengan lintasan gerak partikel, maka partikel akan menempuh litasan setengah lingkaran dan menumbuk detektor sejauh x. Buktikan bahwa massa partikel m adalah

$$m = \frac{B^2 q x^2}{8V}$$

Dimana V adalah beda potensial untuk memberikan kecepatan awal partikel sebelum memasuki medan magnet

TUGAS NO 2

•41 ILW A 13.0 g wire of length $L = 62.0$ cm is suspended by a pair of flexible leads in a uniform magnetic field of magnitude 0.440 T (Fig. 28-40). What are the (a) magnitude and (b) direction (left or right) of the current required to remove the tension in the supporting leads?

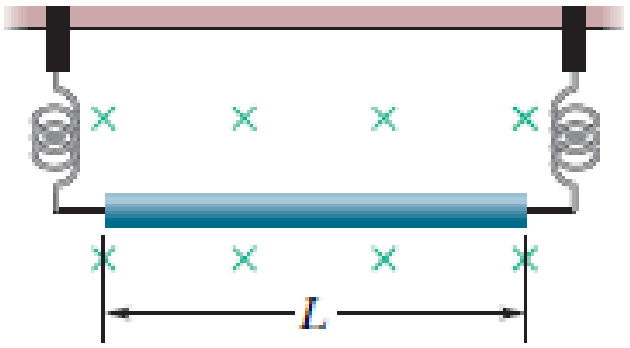


Fig. 28-40 Problem 41.

TUGAS NO 3

••46 In Fig. 28-43, a metal wire of mass $m = 24.1$ mg can slide with negligible friction on two horizontal parallel rails separated by distance $d = 2.56$ cm. The track lies in a vertical uniform magnetic field of magnitude 56.3 mT. At time $t = 0$, device G is connected to the rails, producing a constant current $i = 9.13$ mA in the wire and rails (even as the wire moves). At $t = 61.1$ ms, what are the wire's (a) speed and (b) direction of motion (left or right)?

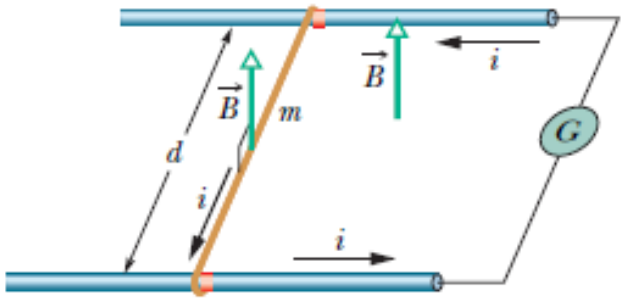


Fig. 28-43 Problem 46.

Tugas no 4

•49 SSM Figure 28-44 shows a rectangular 20-turn coil of wire, of dimensions 10 cm by 5.0 cm. It carries a current of 0.10 A and is hinged along one long side. It is mounted in the xy plane, at angle $\theta = 30°$ to the direction of a uniform magnetic field of magnitude 0.50 T. In unit-vector notation, what is the torque acting on the coil about the hinge line?

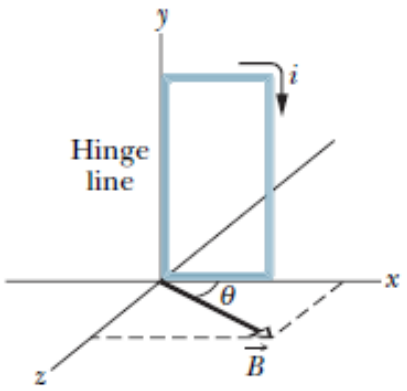


Fig. 28-44 Problem 49.

Tugas no 5

Sebuah motor listrik sederhana mempunya jari-jari kawat loop 2 cm dan banyak ilitan N=50, kemudian diberi arus 1 A, danditepatkan pada meda Magnet 0.01 Tesla.

A. Hitung torsi maksimum
B. Gambarkan grafik torsi terhadap sudut
C. Sebutkan cara yang mungkin dilakukan untuk memperbesar torsi maksimum menjadi dua kalinya